

Kazi Abdul Baset

MSc Robotics & Autonomous Systems | Bachelor of Computer Science Engineering (B.Tech)
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PERSONAL STATEMENT

MSc Robotics & Autonomous Systems student at the University of Sussex, awarded the £10,000 John Kinghorn Scholarship on entry. Started building electronics at age 10 — progressing from RC aircraft & embedded systems to computer vision & full robotic platforms. During my Bachelor I led the suspension department at Team Avitar Motorsports across four national SAE rounds & managed the team's finances, building leadership, organization, & communication alongside technical depth. Currently dissertating on sensor fusion for autonomous train coupling & competing with Sussex Racing in IMechE Formula Student. Looking for graduate engineer roles in robotics, autonomous systems, or electronics.

TECHNICAL SKILLS

Programming: Python || C/C++ || MATLAB || Java || SQL || HTML/CSS

Robotics & Vision: ROS2 || Nav2 || SLAM || Sensor Fusion || Kalman Filtering || OpenCV || 6-DOF Pose Estimation

Embedded & HW: Arduino || ESP32 || PCB Design || Raspberry Pi || RC Drones & Fixed-Wing || Mission Planner

CAD & Simulation: Fusion 360 || SOLIDWORKS || Ansys FEA || FDM/Resin 3D Printing || Rapid Prototyping

LANGUAGES

English (native) || Bengali (native) || Hindi (fluent)

EDUCATION

University of Sussex | Brighton, UK

Oct 2025 – Sept 2026

MSc Robotics & Autonomous Systems

- John Kinghorn Scholar — £10,000 merit award for academic excellence
- Dissertation: Vision & Radar Sensor Fusion for Train Coupling — ~3 cm accuracy over 7–15 m; supports UK/EU Digital Automatic Coupler migration
- Sussex Racing Formula Student — Vehicle Dynamics, Mechanical & Electrical sub-teams

Vellore Institute of Technology (VIT) | Bhopal, India

Jul 2021 – Jun 2025

Bachelor of Technology (B.Tech), Computer Science Engineering — CGPA: 8.43 / 10

- Head of Suspension, Team Avitar Motorsports — full suspension design, simulation, & fabrication across four national SAE rounds; AIR 1 Cost Report at ATVC 2024
- Project developer & lab manager, RAIL Robotics Lab; Head of Events, BitByBit Club

WORK EXPERIENCE

Sussex Racing | University of Sussex

Oct 2025 – Present

Vehicle Dynamics, Mechanical & Electrical Team Member — IMechE Formula Student

- Vehicle dynamics analysis, suspension simulation, & electrical integration for the Formula Student car; collaborated closely with mechanical & electrical sub-teams to meet tight competition deadlines
- Tools: IPG CarMaker, MATLAB, Lotus Suspension Analyser, Ansys

Team Avitar Motorsports | VIT Bhopal

Jul 2021 – Jun 2025

Head of Suspension Department — SAE m-BAJA & ATVC

- Led full suspension design, simulation (SOLIDWORKS), & fabrication; organized team workflow across four national rounds; AIR 1 Cost Report & Overall AIR 14 at ATVC 2024
- Head of Finance (2022–23) — managed team budget, sponsorship negotiations, & cross-departmental communication

Pocket Prints 3D | Kolkata, India

Jan 2023 – Jul 2025

Founder — FDM & Resin 3D Printing Service

- Self-funded home printing business; recovered full setup cost through revenue; organized & led workshops at Kolkata Comic Carnival; official partner at YAA 2.0

RAIL Robotics Lab | VIT Bhopal

Jul 2021 – Jun 2025

Lab Member & Project Developer

- Built & maintained robotics projects; managed lab during open days & university events

PROJECTS

MSc Dissertation: Vision & Radar Sensor Fusion for Train Coupling

2025 – Present

Python || OpenCV || ChArUco || Kalman Filter || PCR mmWave Radar || Raspberry Pi 4

- Bolt-on system providing real-time distance & alignment data for semi-autonomous train coupling — no access to proprietary systems required
- ChArUco 6-DOF pose estimation fused with Doppler mmWave radar; 1D Kalman filter + EMA smoothing + hysteresis state machine

LeaseMate — HackSussex 2026 Overall Grand Prize Winner

Mar 2026

Python || Flask || Tailwind CSS || Local AI || github.com/FrostKazi/LeaseMate

- UK rental contract analyser: audits tenancy agreements against UK housing law (incl. Renters' Rights Act 2025); outputs severity-ranked clause warnings
- Led frontend & backend; designed LLM grounding pipeline with forced structured JSON to prevent hallucinations in a legal context

SpotMicro Quadruped Robot (Bachelor of Technology (B.Tech) Final Year Project)

Jun 2024 – Apr 2025

ESP32-S3 || Python || Fusion 360 || 3D Printing || frostkazi.github.io/SpotMicro/gallery

- Custom 3D printed quadruped robot with web-based remote-control UI & full multi-servo gait implementation built from scratch

Smart Parking System using Computer Vision

Jan – Jun 2023

Python || OpenCV || pytesseract || MySQL || Streamlit

- Automated ANPR on vehicle entry/exit; MySQL billing & logging; live Streamlit dashboard — reduced manual effort ~90%, billing time ~50%

CERTIFICATIONS

- PDRL Certified Drone Professional – PDRL, Apr 2024
- Cloud Computing – NPTEL, Jan – Apr 2024
- Robotic Process Automation — Automation Anywhere, Sep 2023 ([verified credential](#))
- Applied Machine Learning in Python — Coursera, Dec 2022
- The Arduino Platform & C Programming — Coursera, Oct 2021

HOBBIES & INTERESTS

- RC aircraft: 7 fixed-wing designs & 5 quadcopter iterations over 8 years; currently building a multi-axis robotic arm (Fusion 360, ROS, inverse kinematics)
- Published 3D model: RI One RPi4 Console Case — printables.com/model/1417140
- Interests: autonomous robotics, UAVs, motorsport engineering, electronics, PCB design